Penguin Ai

VIA ELECTRONIC SUBMISSION

To: Department of Health and Human Services Assistant Secretary for Technology Policy (ASTP) & Office of the National Coordinator for Health Information Technology (ONC), Attention: HHS Health Sector AI RFI Mary E. Switzer Building, Mail Stop: 7033A 330 C Street S.W. Washington, DC 20201

Re: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care (RIN 0955-AA13)

Dear Secretary Kennedy and Assistant Secretary Posnack,

Penguin Ai appreciates the opportunity to provide comments on the Department's strategy to accelerate the adoption and use of Artificial Intelligence (AI) in clinical care. As a healthcare-native automation platform founded by executives from UnitedHealth Group, Optum, and Kaiser Permanente, we share the Department's commitment to deploying AI that is transparent, equitable, and clinically safe.

We explicitly reject the use of "Black Box" AI in healthcare. Our platform is built on the principle of the "Glass Box" where every AI prediction, from medical coding to clinical appeals, is traceable back to a specific source of truth within the patient record. We believe this standard of Traceability and Governance is essential for building trust among providers and patients.

On the subsequent pages we offer specific responses to questions posed in the RFI, drawing on our experience deploying autonomous "Digital Workers" for healthcare providers.

Sincerely,

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Responses to Specific Questions

1. **What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?**

Response

HHS must address three primary friction points that currently stifle the rapid, secure, and cost-effective deployment of AI:

- **Access to Financial and Clinical Data (EMR & RCM Integration):** The top technical hurdle to AI adoption is the inability to seamlessly integrate with and extract high-quality data from existing Electronic Health Records (EHRs) and Revenue Cycle Management (RCM) systems. Currently, 76% of revenue cycle leaders rate integration between AI tools and core EHR systems as a major or severe challenge. Poor data quality, brittle system interfaces, and a lack of standardized APIs prevent AI models from functioning effectively, starving them of the data needed to generate accurate insights.
- **Capital Constraints & Misaligned Incentives:** Legacy fee-for-service regimes often do not account for the efficiency gains of AI. "Inertia" from these models means that innovations lowering net spending often inadvertently reduce provider revenue. Furthermore, margin pressures (e.g., MLR caps) restrict the available capital for payers to invest in long-term AI infrastructure that may not yield immediate returns.
- **Regulatory Uncertainty & Fragmented Governance:** The absence of a standardized, predictable regulatory framework creates novel legal issues regarding liability, indemnification, and security - particularly for non-medical devices. This ambiguity discourages private sector risk-taking and leads to fragmented adoption standards across the industry.
- **The Operational Skills Gap :** There is a critical deficit in specialized talent capable of managing data ingestion pipelines and model monitoring. Clinicians and administrators often lack the necessary "AI Literacy" to evaluate the robustness and ethical implications of these tools, leading to implementation paralysis.

Recommendations

To mitigate these barriers, HHS should leverage its regulatory and reimbursement pillars to foster a predictable environment:

- **Mandate Administrative Data Interoperability:** To overcome the integration barrier, ASTP/ONC must expand interoperability standards (e.g., USCDI) to explicitly include comprehensive administrative and financial data classes. Certification requirements should enforce standardized API access (like FHIR) to both EHR and RCM systems, ensuring AI tools can securely access necessary data without vendor friction. Details offered in Question 2, Recommendation 4 below.

- **Modernize Reimbursement:** Continue shift away from fee-for-service regimes and include payment policies that reward "high-value" AI interventions. Align federal incentives so that AI is deployed to enhance productivity and reduce burden, rather than just increasing volume.
- **Establish a Predictable Regulatory Posture:** Create a regulatory stance that is well-understood, predictable, and "proportionate to the risks presented." HHS must take an active role in clarifying liability and "safe harbor" frameworks for administrative AI tools to enable rapid innovation. More on this in our response to Question 2, specifically in Recommendations 3 and 5.
- **Support Workforce Readiness:** Fund the development of standardized role-based training and "implementation science" to ensure clinicians and administrators can integrate and oversee high-impact AI systems safely.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

Response

HHS should prioritize programmatic and regulatory shifts that clearly distinguish between *Clinical AI* (diagnostic/treatment) and *Administrative AI* (operational/revenue cycle). To incentivize effective use, it is critical to align financial rewards with the adoption of "high-value" AI that reduces administrative waste, while establishing a regulatory "safe harbor" for transparent, audit-ready administrative tools. By reducing the administrative burden through automation, we return time and energy to clinical care.

Below are specific recommendations regarding programmatic incentives and regulatory revisions.

Part 1: Programmatic Design Changes to Incentivize AI

Recommendation 1: Align MIPS Incentives with AI-Driven Burden Reduction

- **Proposed Change (42 CFR Part 414):** HHS should revisit the Merit-based Incentive Payment System (MIPS), specifically the Improvement Activities performance category (defined in 42 CFR § 414.1305 and § 414.1380).

- **Actions:**
    - Explicitly classify the adoption of "AI-assisted Administrative Automation" (e.g., automated coding, chart abstraction, and prior authorization submission) as a High-Weighted Improvement Activity.
    - Create a specific sub-category for technologies that demonstrate a measurable reduction in "time-to-care" or administrative overhead costs.
- **Rationale:** Current MIPS categories heavily favor clinical data reporting, which often *adds* to provider burden. By financially rewarding the adoption of AI tools that *remove* administrative tasks, HHS can accelerate the deployment of technology that directly combats clinician burnout and operational inefficiency.

Recommendation 2: Transition from Fee-for-Service to Value-Based AI Models

- **Proposed Change (CMMI Authority):** The Center for Medicare and Medicaid Innovation (CMMI) should pilot a "Shared Savings for Administrative Efficiency" model that builds upon the foundational goals of the ACCESS program and the WISeR model.
- **Action:** Instead of reimbursing for AI as a discrete service, allow providers to retain a higher percentage of shared savings in Alternative Payment Models (APMs) if they utilize certified AI agents to reduce the cost of Revenue Cycle Management (RCM) and utilization review.
- **Rationale:** We applaud CMS's ongoing efforts to advance care delivery through programmatic design, but legacy fee-for-service regimes create "inertia" that diminishes the promise of AI. If providers are reimbursed strictly for volume, there is little financial incentive to use AI to become more efficient. Shared savings models that specifically target administrative waste (estimated at 25% of US healthcare spend) will supercharge the objectives of models like WISeR, driving rapid market adoption of high-value AI.

Part 2: Regulatory Revisions to Augment AI Development

Recommendation 3: Clarify Scope to Create a "Presumption of Safety" for Administrative AI

- **Proposed Change (Guidance on HTI-1 & 45 CFR Part 170):** HHS and ASTP/ONC should clear regulatory confusion by establishing a distinct boundary between Predictive Decision Support Interventions (DSI) used for clinical care and Administrative AI used for RCM, coding, and documentation.
- **Action:** Issue guidance explicitly clarifying that AI tools utilized solely for Payment and Health Care Operations (as defined in 45 CFR § 164.501) fall outside the scope of the "Predictive DSI" definition (45 CFR § 170.315(b)(11)) and are therefore not subject to "clinical validity" testing requirements.

- **Standard:** To qualify for this exclusion, the AI must utilize a "Glass Box" / Deterministic Traceability architecture that links every output (e.g., a billing code) back to a specific, human-verified source in the medical record and allows for human override.

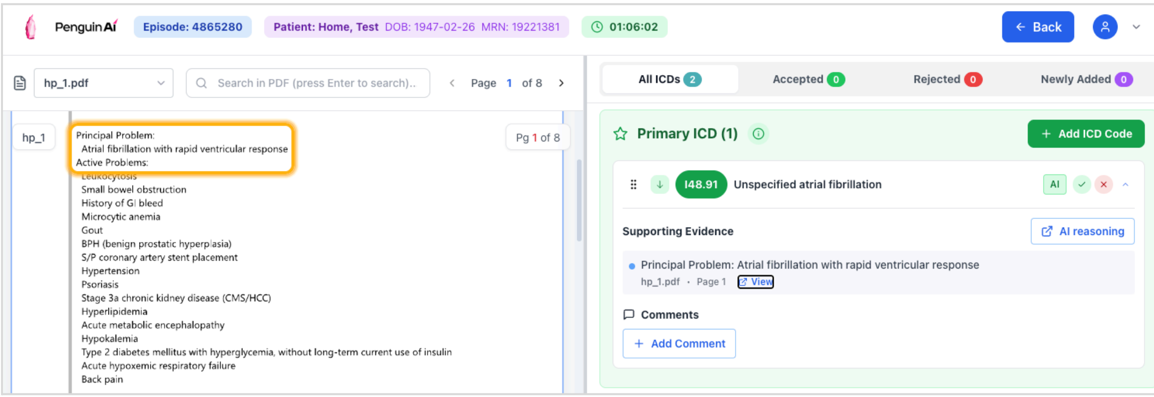

Exhibit A: Example of "Glass Box" Traceability. The AI output (ICD-10 code) is deterministically hyperlinked to the specific, human-verified sentence in the unstructured clinical note.

- **Rationale:** Administrative errors (e.g., a wrong billing code) are fundamentally different from clinical errors (e.g., a missed diagnosis). They are financial, reversible, and auditable. Subjecting operational algorithms to "clinical safety" standards creates regulatory ambiguity that stalls innovation. Clarifying that transparent, administrative tools are *not* "Predictive DSIs" creates a "Safe Harbor" that encourages the market to adopt audit-ready automation immediately.

Recommendation 4: Expand Interoperability Standards for Administrative Data

Proposed Change (45 CFR Part 170 - ONC Health IT Certification): HHS should update the United States Core Data for Interoperability (USCDI) standards to support AI-driven administrative workflows.

- **Action:** Expand USCDI as part of USCDI v7 to include a dedicated "Administrative Automation" domain, or expand the existing “Health Insurance Information” domain.
    - **Specifics:** Standardize data elements for Prior Authorization Criteria, Payer Denial Reasons, and Claim Status metadata. Currently, these often exist as unstructured text or proprietary codes.

Rationale: AI agents require structured data to function deterministically. If a denial reason is unstructured text, the AI must "guess," increasing the risk of hallucinations. By standardizing the "Health Insurance Information" domain, HHS enables AI agents to read the "state" of a claim deterministically, fueling the development of robust, interoperable AI solutions for both clinical and administrative care.

Recommendation 5: Modernize HIPAA for "Privacy-by-Design" AI

Proposed Change (45 CFR Parts 160 and 164 - HIPAA Privacy Rule): HHS should revisit the interpretation of "Minimum Necessary" standards regarding AI training on De-Identified data.

- **Action:** Provide a specific safe harbor for "Privacy-by-Design" AI architectures that utilize automated de-identification and edge-processing to scrub PHI *before* data enters a model training environment.

Rationale: Protecting patient confidentiality is paramount, but ambiguity regarding the "ingestion" of data for AI improvement leads to implementation paralysis. Clear guidance that incentivizes architectural privacy (scrubbing data at the edge) rather than contractual privacy (BAAs alone) will augment the industry's ability to develop safer, more capable AI models.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (*e.g.,* contracts, grants, cooperative agreements, and/or prize competitions)?

Response

HHS identifies AI software that assists in clinical care but does not meet the FDA's definition of a medical device as a critical area for innovation. These administrative and general-purpose tools include:

- **Ambient Scribes**: AI that converts spoken clinician-patient dialogue into clinical notes.
- **Administrative Automators**: Tools for Revenue Cycle Management (RCM), medical coding, or scheduling.
- **Workflow Support**: Software that manages general office operations or serves as a secure messaging platform.

HHS should prioritize evaluation methods that bridge the gap between technical performance and real-world administrative impact, emphasizing operational ROI, human-in-the-loop oversight, and long-term model reliability.

Evaluation Methods for Digital Workers

Evaluation must move beyond simple accuracy to focus on the safety and efficiency of digital workers:

- **Pre-deployment: Validation with Real-World Data (RWD)**: Use secure platforms to validate models against millions of de-identified patient records - including unstructured data - to ensure the AI can handle clinical complexity before it goes live.
- **Post-deployment: Real-Time Performance Monitoring**: Implement continuous monitoring to track KPIs such as "cost-to-collect," denial rates, and time-to-authorization.
- **Iterative Robustness Testing**: Regularly test models for "out-of-distribution" robustness to ensure they do not fail when payer rules or clinical documentation patterns change.

Key Metrics for Clinical Care

Metrics must demonstrate both financial and operational value:

- **Operational Throughput**: Measuring the speed of administrative tasks, such as reducing prior authorization time significantly.
- **Accuracy and Correctness**: Maintaining high rates of clinical correctness (e.g., 95.5%) in specialized models to ensure patient safety.
- **Revenue Integrity**: Tracking the percentage of appeals overturned and the reduction in the A/R backlog.
- **Algorithmic Fairness**: Using subgroup performance metrics to detect and correct bias, ensuring the AI does not discriminate based on demographic data.

Recommendations

To foster a "healthcare-native" approach to AI evaluation and deployment, HHS should implement the following support mechanisms:

Human-Centered Evaluation: The "Human-in-the-Loop" Model

- **Establish Feedback Loops**: HHS should support the development of systems that enable human teams to provide real-time feedback used to retrain AI models and refine workflows.
- **Prioritize Decision Support**: Incentivize AI that "distills and presents" information to human decision-makers, ensuring final clinical and financial decisions remain evidence-based and compassionate.

Impactful HHS Support Mechanisms

- **Cooperative Research and Development Agreements (CRADAs)**: Use CRADAs to develop standardized, de-identified datasets that startups can use for model validation.
- **Grants for Workforce Training**: Provide grants to help providers and payers close the "skills gap" by training staff to manage and audit agentic AI systems.
- **Prize Competitions for Interoperability**: Use prize competitions to incentivize the creation of "system-agnostic" AI tools that plug into existing EHRs and payer portals

without requiring a total system "rip and replace".

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Response

The successful adoption of AI in clinical care is not merely a technical challenge but a governance and leadership imperative. While executive leadership sets the vision, the actual integration of AI into workflows (clinical, financial, operational) depends on a broader circle of stakeholders who manage the intersection of clinical necessity, technical infrastructure, and financial viability.

The Expanded Circle of Influence

While the C-Suite provides the overarching strategy, successful adoption hinges on specialized roles that manage day-to-day operational risks:

- **For Providers (Hospitals & Health Systems):**
    - **Head of Revenue Cycle Management (RCM):** Acts as a primary driver for adoption due to their responsibility for financial health, medical coding efficiency, and billing accuracy.
    - **VP of Revenue Cycle and CFO:** Prioritize AI solutions that deliver measurable ROI by reducing the "cost-to-collect" and minimizing claim denials.
    - **Director of Clinician Operations:** Crucial for ensuring that AI tools provide genuine relief for documentation burdens rather than introducing new administrative tasks.
- **For Payers (Insurance Companies):**
    - **VP of Provider Network Strategy:** Influences the adoption of tools that streamline data exchange between payers and providers.
    - **VP of Utilization Management & CMO:** Focus on AI-driven prior authorization reviews to ensure medical necessity and accelerate care delivery.
    - **Director of Regulatory Compliance:** Exercises significant oversight to ensure that AI data ingestion meets strict HIPAA and civil rights standards.

Recommendations

To address the persistent administrative friction points that stall AI adoption, HHS should prioritize the following areas:

Overcoming Primary Administrative Hurdles

HHS must take an active role in mitigating the "non-technical" barriers to innovation:

- **Mitigate the "Skills Gap":** Provide support for healthcare organizations to build internal expertise so they are not solely reliant on vendors, fostering a culture of trust and effective monitoring.
- **Relieve Budgetary Constraints:**
  - **Payers:** Address the limitations posed by Medical Loss Ratio (MLR) growth to allow for necessary technology investments.
  - **Providers:** Create financial pathways to offset high upfront costs and "rip and replace" infrastructure projects in an environment of reduced payment rates.
- **Address Governance and Liability "Inertia":**
  - **Clear Liability Guidelines:** Establish a legal framework that clarifies responsibility for AI errors (clinician vs. vendor) to eliminate the "wait and see" culture.
  - **Workflow Transformation Support:** Provide guidance for transitioning from manual, legacy workflows to "agentic" systems, recognizing the significant change management required.
- **Standardize AI Data Provenance and Payloads:** While AI models are highly capable of extracting data from unstructured sources (faxes, PDFs, voice), the technical bottleneck is integrating that output into core systems. In other words, the receiving EHR system doesn't speak English; it speaks in discrete codes (SNOMED, ICD-10). If every AI vendor reads a .pdf and maps "heart attack" to a different code, or outputs the data in a proprietary JSON format, the hospital's IT team has to build custom "plumbing" for every single AI tool to get those findings back into the EHR. ASTP should invest in standards (e.g., expanding FHIR Provenance resources) that standardize how AI-extracted data, confidence scores, and source-document traceability are communicated to EHRs, enabling plug-and-play interoperability without custom mapping.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Response

Penguin Ai commends ASTP/ONC and CMS for their monumental work in establishing the current Interoperability Framework. Interoperability standards (like FHIR and USCDI) have succeeded in unlocking "read" access to patient records, laying the critical groundwork for modern health IT. However, the market remains constrained by a severe asymmetry in "write" access. Large Electronic Health Record (EHR) and Revenue Cycle Management (RCM) vendors maintain technical and contractual moats that restrict third-party AI from pushing insights directly into native workflows.

This "Pull vs. Push" asymmetry relegates new AI vendors to use of "read-only" data supporting workflows, dashboards or side-car applications. The resulting workflow fragmentation forces clinicians and coders into "swivel-chair" processes that increase cognitive load and hinder the adoption of even the most accurate AI models.

To widen market opportunities, fuel research, and accelerate AI development [and the path toward seamless cross-system workflows] built upon the foundation of the Interoperability Framework, the focus must shift from basic data access to establishing standards for seamless "write-back" capabilities, standardizing administrative data, and creating objective benchmarking tools.

Recommendations

1. Mandate "Write-Back" Standards for Native Workflow Integration

- **Action:** ASTP should prioritize certification criteria (e.g., in upcoming HTI rulemakings) that mandate standardized, bi-directional API capabilities for both clinical and administrative workflows. This must include policies that limit the fees legacy vendors can charge third parties for "write" access.
- **Rationale:** Accelerating AI requires the ability to push insights, such as an AI-generated Prior Authorization payload, an extracted ICD-10 code, or a pre-visit clinical briefing, directly into the user's native screen. Without affordable, standardized write-back capabilities, AI cannot achieve its primary goal of reducing administrative burden.

2. Expand USCDI to Include Structured Administrative Data

- **Action:** As requested in Question 2, Recommendation 4, widen the USCDI and USCDI+ frameworks to formally standardize the exchange of administrative and financial data (e.g., Claims Data, Payer Denial Reasons, Prior Authorization Criteria, Claim Status).

- **Rationale:** AI development is currently bogged down by the need to parse unstructured faxes, PDFs, and legacy X12 EDI formats. Standardizing these administrative elements allows AI to function deterministically across different EHRs and RCMs, expanding market opportunities beyond clinical diagnostic AI into high-value back-office automation.

3. Establish "Glass Box" Provenance Standards

- **Action:** ASTP should establish FHIR Provenance standards specifically for AI-generated data. Any AI insight pushed into a certified health IT system must include metadata linking the output directly to its human-verified source (e.g., a coordinate mapping that highlights the exact sentence in a clinical note justifying a suggested code).
- **Rationale:** Trust is a major barrier to adoption. If an AI writes data into an EHR, downstream users need to know exactly how that conclusion was reached. Standardizing AI provenance allows for seamless "Human-in-the-Loop" review without disrupting the workflow.

4. Develop Open-Source "Gold Standard" Benchmarking Datasets

- **Action:** HHS should support the creation of open-source, de-identified datasets that combine longitudinal clinical records with claims and payment data.
- **Rationale:** Research and market competition are currently limited by a lack of objective benchmarking tools outside of proprietary EHR silos. Developers need access to diverse, standardized datasets to evaluate the clinical robustness and the financial "cost-to-collect" impact of their AI models, ensuring safer and more effective solutions.

Conclusion

We applaud ASTP and ONC for leading this conversation. The future of healthcare AI lies not in "magic box" solutions, but in transparent, auditable, and healthcare-native systems that reduce burden and improve outcomes.

We welcome the opportunity to discuss our "Glass Box" architecture and Governance frameworks further with your team.